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Zixu Wang

PhD Student • Lancaster University

Profile

Full-time PhD student at Lancaster University researching quantum-inspired agentic AI for financial modelling. Python and PyTorch experience spans data preparation, model training, evaluation and explainability across medical imaging, ECG and LLM projects. First author of a PRCV 2026 paper, seeking AI/ML engineering opportunities.

Education

Lancaster University — Lancaster, UK
PhD (full-time) — **Oct 2026 – Present**
Supervisor: Dr. Richard Jiang. Research: quantum-inspired neural agentic AI for autonomous financial modelling, focusing on tensor networks, reinforcement learning and explainable decision-making.

Lancaster University — Lancaster, UK
MSc Artificial Intelligence — **Oct 2025 – Sep 2026**
Expected Distinction; formal award pending. Dissertation: 75/100.

Aston University — Birmingham, UK
BSc Computer Science (Hons) — **Sep 2021 – Jul 2024**
Second Class Honours, Lower Division. Mathematics: 79/100; Human–Computer Interaction: 70/100.

Teaching Experience

Teaching Assistant, SCCM7450 · Lancaster University — *Oct 2026 – Present*

- Support laboratory sessions, answer student questions, and mark and grade coursework.

Publication

Zixu Wang, Richard Jiang. **How Far Can a Single Lead Go? A Construct-Validity Audit of ICD-Based ECG Benchmarks on 54,357 MIMIC-IV Patients.** *PRCV 2026.*

- Benchmarked six model families with patient-level splits; audited ICD-derived labels using no-ECG baselines, age stratification, label sensitivity and zero-shot transfer to PTB-XL.
- Identified demographic and care-pathway contributions to aggregate AUC (no-ECG: 0.816; best ECG: 0.800), with ECG gains of 0.05–0.07 within age strata 50–79.

Selected Projects

MSc Dissertation • Region-Level Explainability on PD-DBS Facial Images — *Jun – Sep 2026*
Explainable Face2Gene Diagnosis project, Lancaster University | Supervisor: Dr. Richard Jiang

- Designed an eight-region occlusion audit and Anatomical Evidence Vector (AEV) pipeline on 2,343 grayscale 32×32 images, quantifying model sensitivity for pre-/post-DBS classification.
- Evaluated NumPy MLP and compact CNN models with calibration, similarity filtering, spatial permutation, region-only tests, atlas geometry variants and Grad-CAM comparisons.
- Four regional contrasts persisted after confidence matching and FDR correction; right cheek/zygomatic ranked first by balanced accuracy in 10 of 11 atlas variants.

Undergraduate Final-Year Project • Restaurant Management System — *Aston University*
BSc Computer Science (Hons) | Individual project

- Developed a restaurant back-office management system, covering requirements analysis, implementation, testing and delivery.

Technical Skills

- **Programming:** Python, SQL, Java
- **AI / ML:** PyTorch, scikit-learn, NumPy MLPs, CNNs, transfer learning, clustering, ECG analysis
- **NLP / LLMs:** Text preprocessing, OpenAI / Anthropic / Google API integration, agent simulations
- **Data:** pandas, NumPy, matplotlib, seaborn, NetworkX
- **XAI / Evaluation:** Occlusion, Grad-CAM, calibration, AUROC/AUPRC, FDR
- **Tools & Languages:** Git, Linux, Jupyter, LaTeX; Chinese (native), English (fluent)